

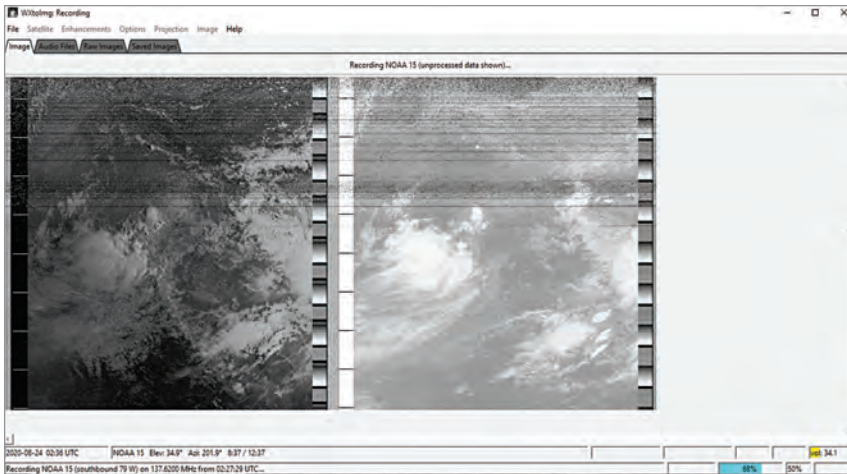


Fig. 15: Screenshot of WXtoImg when a satellite image is being received

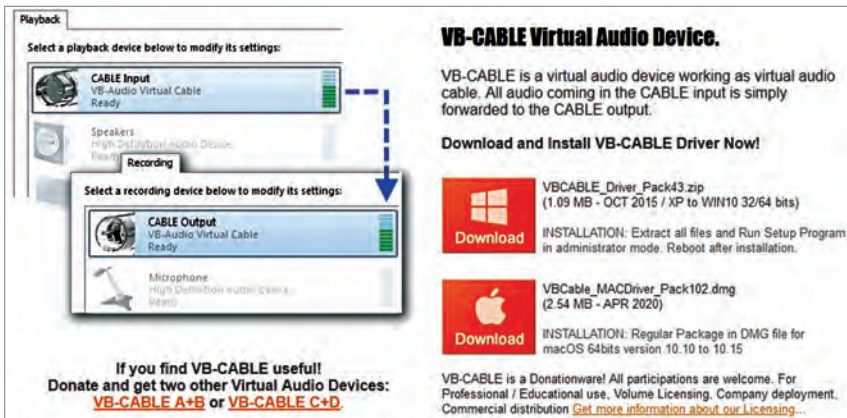


Fig. 16: The virtual audio cable software

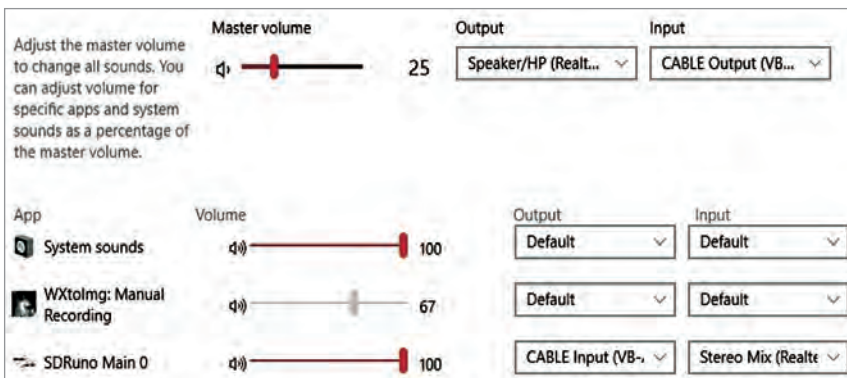


Fig. 17: The system sound settings and connections from the receiver output to virtual audio cable

TABLE 2
FREWARE SOFTWARE REQUIRED FOR THIS PROJECT

Functionality	Suggested Option	Website
APT Satellite Decoding	WXtoImg	https://wxtoimgrestored.xyz/
Software/Virtual Audio Connections	VB-Cable	https://www.vb-audio.com/index.htm
Satellite Pass/Trajectory Prediction	Orbitron GPredict	http://www.stoff.pl/ http://gpredict.oz9aec.net/

(SDR) available as USB dongles. These not only offer reception of the broadcast AM/FM bands but a wide coverage from the LF/MF to frequencies as high as 1.7 to 2GHz, depending upon the device. One commonly used is the RTL-SDR RTL2832-Ver3 USB dongle that can be purchased from e-commerce websites. Alternately, there is RSP1A from SDRPlay, but at a higher price tag.

The RTL-SDR dongle uses 8-bit ADC while the RSP1A has a powerful 14-bit ADC. The performance differs, so does the pricing. It differs largely! Therefore, it is up to you to select as per your budget. The author uses both—RSP1A for the NOAA satellites and RTL-SDR for the Meteor-M2. These USB SDRs come with their own user interface, but that only allows you to operate the radio in a conventional manner like tuning to a particular frequency, setting receive mode such as AM, FM, DSB, CW or USB/LSB in case of single side band reception and bandwidth controls, etc. See Fig. 12 for receiver hardware/devices.

You can explore the software of your choice with the help of the user guide. We shall here focus more on utilising this radio (SDR) to receive the weather satellite signals and use related software tools to decode the images. The screenshots of the GUI-dashboards of these SDR receivers are shown in Fig. 13 and Fig. 14. You can get acquainted with the software by receiving standard broadcasts before proceeding further for the weather satellite reception as there are other set of software (as explained later in the text) required for decoding the images and converting the data into image files.

Software. Besides the SDR software, you shall require a few more, as summarised in Table 2.

Decoding software. The most popular and easily available as freeware is the WXtoImg software for decoding APT transmissions. This software not only records the audio received from